

1 What should I submit, where should I submit and by when?

Your submission for this minor assignment will be one PDF (.pdf) file only. Instructions on how to prepare and submit this file are given below.

Assignment Package:

<https://www.cse.iitk.ac.in/users/purushot/courses/ml/2025-26-w/material/assignments/minor3.zip>

Deadline for all submissions: May 02, 2026, 9:59PM IST

Code Submission: *None needed – do not submit any code (ipynb, py, etc) files for this minor assignment*

Report Submission: on Gradescope

There is no provision for “late submission” for this minor assignment.

1.1 How to submit the PDF report file

1. The PDF file must be submitted using Gradescope in the *group submission mode*.
2. Note that unregistered auditors cannot make submissions to this minor assignment.
3. Make only one submission per assignment group on Gradescope, not one submission per student. Gradescope allows you to submit in groups - please use this feature to make a group submission.
4. Link all group members in your group submission. If you miss out on a group member while submitting, Gradescope will think that person never submitted anything.
5. You may overwrite your group’s submission as many times as you want before the deadline (submitting again on Gradescope simply overwrites the old submission).
6. Do not submit Microsoft Word or text files. Prepare your report in PDF using the style file we have provided (instructions on formatting given later).

Problem 2.1 (Melbo’s ‘Mazing Mixtures’). Melbo wants to explore the power of mixture models to better predict pollutions levels using low-cost sensor data and contribute towards cleaner air in Indian cities. For this task, you need to refer to the following resources:

1. Calculations and code for the lecture “The EM Algorithm” available at <https://www.cse.iitk.ac.in/users/purushot/courses/ml/2025-26-w/material/lectures/lec16.zip>
2. The AQI dataset available at [https://github.com/purushottamkar/aqi-satvam/blob/main/data/DD1\(Oct\).csv](https://github.com/purushottamkar/aqi-satvam/blob/main/data/DD1(Oct).csv)

(60 marks)

Your Task. The following enumerates 5 parts to the question. All parts need to be answered in the PDF file containing your report. **Do not submit any code files for this minor assignment.** Thus, submit only a single PDF file as your submission. The lecture code uses a mixture model to solve a regression problem with a single real-valued output (Ref. O3(ppb)) and 5 features (Time, Temp(C), RH(%), o3op1(mV), o3op2(mV)).

Note that for all parts below, you must use the same train-test split as used in the lecture code i.e. first 80% valid time-stamps as train and rest as test. Do not use random or other train-test split. Do not change the features, i.e. do not use Ref. NO2(ppb) or no2op2(mV) or no2op2(mV) as features. You may however, make changes to data processing e.g. you may make changes to how training data is pre-processed e.g. feature normalization, converting timestamp to hour-of-day, changing the regularization constant alpha in linear_model.Ridge etc. You may also optimize how prediction is done e.g., the lecture code is inefficient as it gets predictions for every data point from every model in the ensemble, etc. Also note that in parts 1 and 2 below, your test MAE must be obtained by first training a classification model to predict the best regression model i.e. your test MAE should not be *unrealistically good* (e.g. 2.95 in the lecture code) but obtained via the auxiliary model (e.g. 9.16 in the lecture code).

1. Melbo wishes to find how many iterations of the EM algorithm are sufficient for good test performance. Keeping `n_components = 4` i.e., the same as in the lecture code, repeat the experiment after changing the `n_iter` parameter for the function `doEMMR()` and plot a curve with `n_iter` in x-axis and test MAE as y-axis. Vary `n_iter` from small values e.g. `n_iter = 1` all the way to large values e.g. `n_iter = 100`. Try at least 10 different values of `n_iter` in your plot. Discuss in what range of `n_iter` is there large variation in test performance and after what value of `n_iter` is there no/little change to test performance if increasing `n_iter` (the *knee point*). (10 marks)
2. Melbo now wishes to find out how many components are needed in the EM algorithm for good test performance. Repeat the experiment after changing the `n_components` and plot a curve with `n_components` in x-axis and test MAE as y-axis. Try (at least) the following values for your plot `n_components = {1, 2, 4, 8, 16, 32}`. Keep `n_iter` fixed to the knee point you identified in part 1 above. (10 marks)
3. For experiments in part 2 that used `n_components < 10`, comment on what seems to distinguish the components from each other, e.g., in the lecture code, it seemed that with `n_components = 4` the components seemed to separate timestamps with low/medium/high values of O3. What happens when `n_components` is smaller/larger than 4? (10 marks)
4. Fix `n_components = 4` and repeat the experiment from part 2 five times, each time excluding one of five features. Thus, in the first experiment, train/test the model with only (Temp(C), RH(%), o3op1(mV), o3op2(mV)) as features i.e. Time is excluded as a feature during both train and test. In the second experiment, train/test the model with only (Time, RH(%), o3op1(mV), o3op2(mV)) as features i.e. Temp(C) is excluded as a feature during both train and test, and so on. Keep all other hyperparameters e.g. `n_iter`, `alpha` etc fixed to the values you used for them in part 2. Plot the same graph as you plotted in part 2 except this time, it would have six curves, five corresponding to each of the five experiments corresponding to each excluded feature, and the sixth curve being the one you plotted in part 2 with all features included. Comment on which feature seems to contribute most/least to the models prediction by inspecting how much does performance drop if that feature is excluded. (10 marks)

5. Melbae pointed out to Melbo that mixed regression, that learns multiple regression models and a classification model to send each data point to one (hopefully the best) regression model, effectively implements a two-layer regression tree with the root node and as many leaves as there are components in the mixture. The root's children are all leaves and the root sends data points to one of its children. Melbo found this connection intriguing and wondered if it would be better to directly use a regression tree to solve the problem instead of running the EM algorithm. To clarify things for Melbo, use the `sklearn.tree.DecisionTreeRegressor` model to solve this problem. Use all 5 features (Time, Temp(C), RH(%), o3op1(mV), o3op2(mV)), (tune the `criterion` and `splitter` and other hyperparameters – report their best values). Train the `DecisionTreeRegressor` several times, each time changing the `max_depth` hyperparameter. Explore at least 10 values of this hyperparameter ranging from small (e.g. `max_depth = 2`) all the way to large (e.g. `max_depth = 10`). After running all experiments, plot a curve with x-axis as total prediction time over the entire test set, and y axis as test MAE. Show two curves on this plot, one corresponding to the various decision trees you learnt by changing `max_depth` and the second curve corresponding to the various mixture models with different number of `n_components`. This curve will show which model offers better performance w.r.t. speed of test predictions. Discuss your findings. (20 marks)

Using Internet Resources. You are allowed to refer to textbooks, internet sources, research papers to find out more about this problem and for specific derivations e.g. the arbiter-PUF problem. However, if you do use any such resource, cite it in your PDF file. There is no penalty for using external resources with attribution but claiming someone else's work (e.g. a book or a research paper) as one's own work without crediting the original author will attract penalties.

Restrictions on Code Usage. You should not use any library other than what is already imported in the Jupyter notebook in the assignment package (you do not need to anyway). Since this minor assignment does not require any code submission, we will trust you on this one. However, you may use fancier libraries like seaborn or plotly to plot graphs etc.

2 How to Prepare the PDF File

Use the following style file to prepare your report. The ZIP file contains an example file and instructions.

<https://media.neurips.cc/Conferences/NeurIPS2025/Styles.zip>

You must use the following command in the preamble

```
\usepackage[preprint]{neurips_2025}
```

instead of `\usepackage[preprint]{neurips_2025}`. Use proper $\text{\LaTeX}$ commands to neatly typeset your responses to the various parts of the problem. Use neat math expressions to typeset your derivations. Remember that all parts of the question need to be answered in the PDF file. All plots must be generated electronically - hand-drawn plots are unacceptable. All plots must have axes titles and a legend indicating what are the plotted quantities. Insert the plot into the PDF file using $\text{\LaTeX}$ `\includegraphics` commands.